Multiview Prototypical Learning and Semantic Explaining Transformer for Hyperspectral Change Detection

Yixiang Huang, Yi Cen, Ruoxi Song, Wenchao Qi, Changping Huang, and Lifu Zhang, *Senior Member, IEEE*

***Abstract*—Abundant hyperspectral remote sensing data provides change detection (CD) technology with the opportunity to accurately distinguish land cover transformations across a consistent area over time. However, most deep learning methods for hyperspectral CD are performed in a fully supervised manner, which requires a large number of pixel-level labeled samples and is highly time-consuming. Therefore, this study combines semi-supervised learning (SSL) to reduce label acquisition costs. However, there are many challenging areas, such as low contrast regions, which increase the extra burden to interpret general semantic information from land cover attributes. To address these challenges, we introduce an innovative multiview prototype learning (MVPL)-based general semantic explaining temporal–spatial–spectral transformer (SETSST) for hyperspectral CD: 1) first, we design self-view and cross-view prototypical similarity maps according to distinct perturbed views of input samples, which ensures appropriate feature interactions by leveraging the low-density aggregated representation; 2) second, the semi-supervised pretraining framework based on weak-to-strong prediction alignment and MVPL is designed to simultaneously capture features from both labeled and unlabeled samples; and 3) to comprehensively interpret semantic explaining information in real land cover attributes, we propose the semantic explaining change analysis (SECA) module. This module enhances feature explanation by clustering them into several change types and an unchanged type, while accounting for the influence of distance to the central pixel. Extensive experiments conducted on three real datasets demonstrate the effectiveness of the proposed MVPL-based semi-supervised pretraining framework and the SECA module in leveraging both labeled and unlabeled samples for hyperspectral CD. This method surpasses the performance of many existing hyperspectral CD approaches and significantly reduces the need for labeled training samples.**

***Index Terms*—Change detection (CD), hyperspectral image (HSI), prototype learning, semi-supervised learning (SSL), transformer.**

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I. Introduction

CHANGE detection (CD) in remote sensing identifies land cover changes within the same area by analyzing images taken at different times [1]. It is widely used in fields such as farmland inspection [2], forest carbon assessment [3], and ice shelf monitoring [4]. Hyperspectral remote sensing, with its ability to capture detailed spectral variations, enhances CD by detecting subtle changes in land cover [5], [6], [7], [8]. Hyperspectral CD involves three stages: preprocessing to improve data quality, applying specialized algorithms to generate change maps, and evaluating performance for accuracy and reliability. CD methods are vital for understanding and monitoring environmental and geographical transformations.

Due to the time-consuming nature of manual interpretation from multitemporal hyperspectral images (HSIs), automatic hyperspectral CD algorithms are continually being developed. These algorithms can be divided into supervised CD and nonsupervised CD (including unsupervised, semi-supervised, and contrastive learning, etc.) based on the quantity of labeled samples required.

In the early stage, most traditional methods belong to the nonsupervised CD category. These include methods based on image comparison operators, such as change vector analysis (CVA) and distance measures [9], [10], postclassification assistant algorithms [11], transformation-based methods such as principal component analysis (PCA) and band selection techniques [12], [13], [14], [15], and independent image classification methods [16], [17]. However, these approaches usually leverage manual features or shallow features, which are vulnerable to error propagation, potentially affecting the accuracy of the detection results.

Supervised hyperspectral CD has become mainstream with the rise of deep learning methods. These models are divided into convolution-based and Transformer-based approaches based on their feature perception capabilities. Convolution methods (e.g., 1-D [18], 2-D [19], [20], [21], and 3-D convolutions [22], [23]) primarily extract features from spectral and

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spatial dimensions but fail to emphasize regions of interest or capture long-range temporal–spatial–spectral relationships. Emerging Transformer-based methods show promise in hyperspectral CD [24], [25], [26], but still face challenges in leveraging the semantic information of ground quality to enhance detection performance.

To reduce labeling costs while leveraging deep learning models, semi-supervised and prototypical learning can be explored for hyperspectral CD. Semi-supervised learning (SSL), based on the weak-to-strong consistency principle, uses limited labeled samples to utilize unlabeled data effectively. For example, feature-match based methods, such as the Match-based framework [27], [45], use pretrained Transformers for feature extraction from unlabeled data, while Liu et al. [28] introduced a multilayer screening strategy for active learning to address label scarcity. Prototypical networks cluster points around prototypes for each class [29], enhancing semantic diversity in predictions by leveraging multiple inputs [30]. This approach can reduce the impact of labeling noise on low-contrast regions. However, such concepts have not yet been tailored for the specific challenges of hyperspectral CD, such as extreme spectral dimensionality. Specifically, existing semi-supervised frameworks for HSCD primarily rely on individual pixel-level or patch-level consistency. They often fail to aggregate global semantic cues from the entire batch, which is particularly detrimental in hyperspectral data where spectral variability can easily lead to pseudo-consistency in local representations. This gap highlights the necessity of a prototypical approach that can capture more stable, class-level semantic anchors. Overall, existing semi-supervised CD methods often overlook the inherent self- and cross-view relationships among perturbed unlabeled samples. This underscores the necessity of enhancing semantic compactness within individual views while enforcing cross-view consistency across different perturbations. Such a mechanism ensures that the model learns invariant representations, effectively bridging the gap between weak and strong augmentations in consistency-based SSL.

This article addresses the hyperspectral CD challenges by utilizing prototypical learning and semantic analysis. However, unlike natural image analysis, the sample awareness changes from image level to patch level, which brings the problem of how to determine the prototype. In addition, the cluster that is used to compute the prototype is changed from a set of class-oriented images to a group of perturbation-oriented random patches, as well. These issues make it more difficult for prototypical learning-based hyperspectral CD.

Therefore, this study aims to develop a prototypical learning framework with a semantic change analysis module for hyperspectral CD. We improve prototypical learning by proposing the multiview prototype learning (MVPL) loss function, incorporating self-view and cross-view similarity maps tailored for hyperspectral tasks. Integrated with a semi-supervised consistency training schedule, MVPL aggregates features from perturbed samples, enhancing prediction diversity while maintaining weak-to-strong consistency. Based on the prior knowledge that real ground scene in the data covers multiple classic change types, we propose a general semantic explaining change analysis (SECA) module. This module categorizes various real ground feature attributes into several change types and one unchanged type. Moreover, the SECA module takes the spatial distance semantic aggregation of the central pixel into consideration. By rearranging semantic change features, semi-supervised MVPL can provide refined and necessary features from both labeled and unlabeled samples. The main contributions of this article can be summarized as follows.

1) We propose MVPL, which includes two distinct feature aggregation prototypes corresponding to different perturbations, to generate self-view and cross-view prototypical similarity maps. By leveraging the low-density aggregated representation that captures common characteristics, this approach ensures appropriate feature interactions among various types of perturbed information.
2) To further maintain prediction alignment, the semi-supervised pretraining framework integrated with the weak-to-strong consistency principle is designed to cooperate with MVPL, which can effectively leverage limited training samples and explore abundant features simultaneously from both labeled and unlabeled samples.
3) To restructure semantic features using prior knowledge of ground object attributes, we developed the SECA module. This module categorizes features into several change types and one unchanged type. In addition, it accounts for the influence of semantic information from central pixel distance aggregation, ultimately enhancing model explainability.

II. Related Work

A. Semi-Supervised Learning

SSL is well-suited for scenarios where unlabeled data is abundant but labeled data is scarce, costly, and time-consuming to collect [31]. SSL relies on assumptions like smoothness, cluster, and manifold to align the problem structure with the model's design. Match series methods, leveraging weak-to-strong consistency under the cluster assumption, effectively utilize unlabeled samples and are widely adopted [27], [32], [33]. In HSI processing, SSL addresses labeling challenges by providing sufficient training samples and extracting features effectively [34]. In hyperspectral classification, SSL addresses imbalanced data challenges by leveraging spatial neighborhood information [35], among which traditional methods often rely on human interventions, such as active learning [36], discriminative random field [37], and band selection [38]. In hyperspectral CD, semi-supervised training often employs pseudo-labeling techniques, using methods like CVA [39] or multilayer cascade screening [28] to generate an initial change map. Despite these advancements, most studies overlook proper feature interaction across perturbed information and fail to maintain disagreement within a semi-supervised consistency constraint framework. Addressing these gaps is crucial for ensuring robust feature extraction and enhancing SSL's effectiveness in hyperspectral applications.

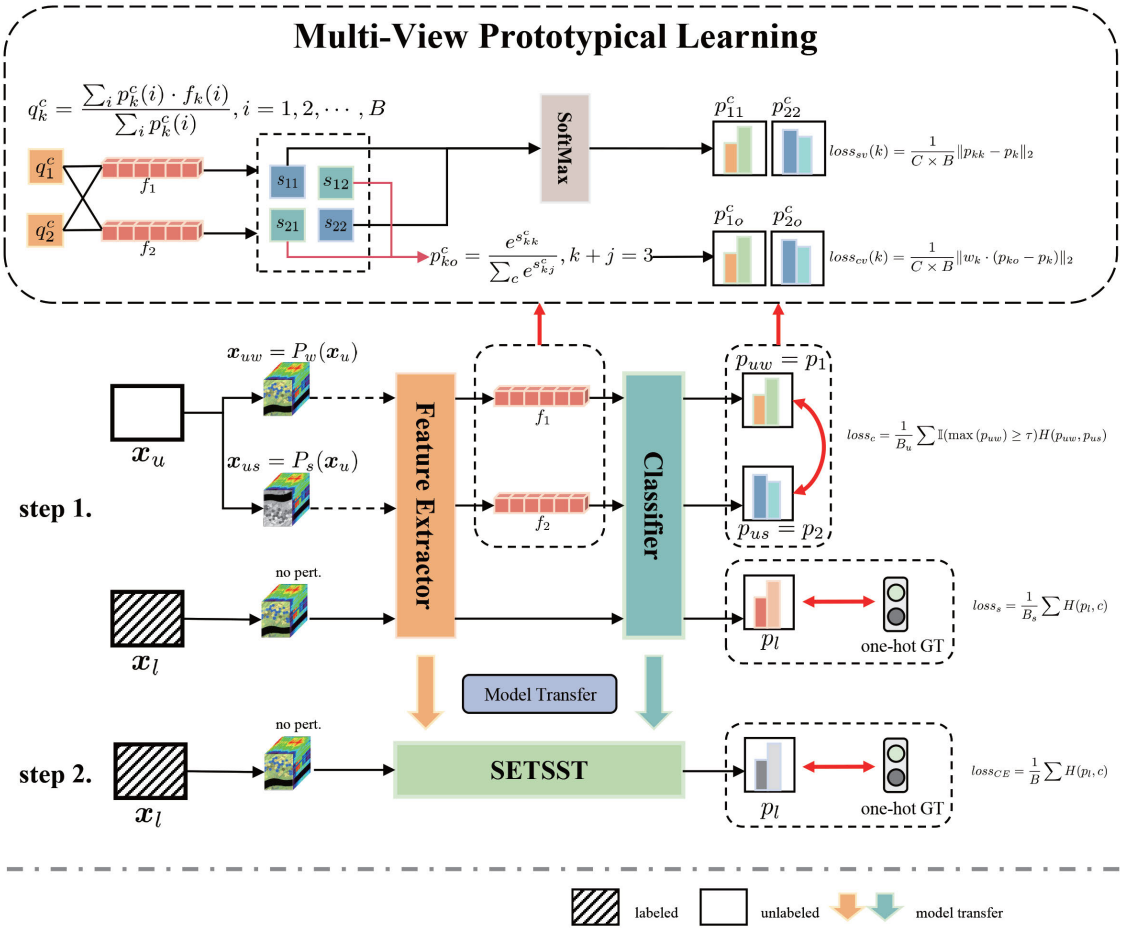


Fig. 1. Overall architecture of semi-supervised multiview prototypical learning.

B. Transformers in Hyperspectral CD

The Transformer model, originally developed for natural language processing, uses an attention mechanism to effectively handle sequence-to-sequence tasks by aggregating contextual information from input sequences [40]. In HSI CD, it filters irrelevant spectral and spatial details [41]. Wang et al. [42] introduce SSA-SiamNet, a model that leverages the similarity between different spectral channels and spatial locations in multitemporal images to adaptively stress the informative contributions of specific channels and locations. To mitigate the impact of patch size on large volumes of spectral and spatial information, Gong et al. [43] have developed adaptive spectral and spatial attention modules that utilize Gaussian distributions. Luo et al. [44] propose a skip-layer attention module based on CBAM to learn the refined spatial–spectral features adaptively. The module is embedded into the skip connections to enhance the regions of interest for the CD task. Moreover, some methods incorporate long-range dependencies in the temporal domain to enhance feature interactions across different time points, such as [24] and [25]. Although Transformer-based methods have discussed the potential of leveraging spatial, spectral, and temporal dependencies, they are still in the early stages and cannot logically restrain semantic features that are based on universal ground change types.

III. Proposed Method

To address challenges in multitemporal HSIs, we propose a MVPL framework with a feature extraction model termed the semantic explaining temporal–spatial–spectral transformer (SETSST). While the backbone architecture inherits the temporal–spatial–spectral attention mechanism from our prior work [45] for robust long-range dependency modeling, the contribution of this study lies in the proposed MVPL framework and the plug-and-play SECA module for semantic-explained CD. Building on that, this brand-new framework can ensure semantic feature interaction while maintaining disagreement under semi-supervised consistency constraints. The training process includes two steps: pretraining with abundant unlabeled and limited labeled samples, followed by fine-tuning. The architecture of the MVPL framework and SETSST is illustrated in Fig. 1.

In this section, we provide a detailed explanation of our approach, covering the overall framework, MVPL (which includes the self-cross prototypical prediction and unsupervised consistency constraint), as well as the SECA module.

A. Self-Cross Prototypical Prediction

In segmentation, a prototype refers to an aggregated representation that encapsulates the shared characteristics of pixel-level features from a specific object or class [30]. In

the context of hyperspectral CD, we specifically extend the concept of a prototype to represent batch-wide, patch-based sample-level features from a particular perturbed view. Here, we give definitions of perturbed views, which are the weakly perturbed view $x_{uw} = P_w(x_u)$ and the strongly perturbed view $x_{us} = P_s(x_u)$. Specifically, a weak perturbation consists of random horizontal flips and rotations to introduce minor adjustments to the original hyperspectral patches. In contrast, strong perturbation incorporates a broader range of transformations, including random horizontal and vertical flips, rotations, and Gaussian noise. Crucially, each perturbation method applies a composition of multiple augmentations from its respective set in a sequential manner, rather than selecting a single operation.

In this work, a "view" is defined as an augmentation state of the same bi-temporal hyperspectral patch used for CD. Self-view refers to feature-prototype matching performed within the same perturbation level (e.g., matching features f_k with prototypes q_k from the same view), encouraging intraview semantic compactness. Cross-view refers to feature-prototype matching across different perturbation levels (e.g., matching features f_k with prototypes q_j aggregated from the other view, $j \neq k$), encouraging interview distribution alignment and semantic information exchange.

Let $p_k^c(i) \in \mathbf{R}$ be the model prediction probability of sample i that belongs to kth view and cth class, $f_k(i) \in \mathbf{R}^D$ refers to the feature representation vector of sample i that belongs to kth view. Therefore, we can define the prototype aggregation within a batch size B_u

$$q_k^c = \frac{\sum_i p_k^c(i) \cdot f_k(i)}{\sum_i p_k^c(i)} \in \mathbf{R}^D, i = 1, 2, \ldots, B_u \tag{1}$$

where D denotes the dimension length of the feature representation. Since the computation of prototypes is based on views with different perturbations, a mini-batch sample is supposed to contain $K \times C$ prototypes. According to calculated prototypes, we can define the self-view similarity via $q_k^c \in \mathbf{R}^D$ and all feature representation vectors $f_k \in \mathbf{R}^{B_u \times D}$

$$S_{kk}^c = \frac{f_k \cdot q_k^c}{\|f_k\| \cdot \|q_k^c\|} \in \mathbf{R}^{B_u}. \tag{2}$$

When consider all C classes, the $S_{kk} \in \mathbf{R}^{B_u \times C}$ can be obtained. Then, the self-view prototypical prediction can be activated by the SoftMax function

$$p_{kk} = F_{SoftMax}(S_{kk}) \in \mathbf{R}^{B_u \times C}. \tag{3}$$

Since the q_k^c is the aggregated prototype in the kth view, it can obtain more homologous features with the corresponding feature map f_k, which can ensure the intraview consistency of prediction. Inspired by this, we can define the corresponding cross-view prototypical prediction. Here, the cross similarity based on cosine similarity can be defined as

$$S_{kj}^c = \frac{f_k \cdot q_j^c}{\|f_k\| \cdot \|q_j^c\|} \in \mathbf{R}^{B_u}, k = 1, 2, k + j = 3. \tag{4}$$

This step ensures that features are linked and that information is exchanged across views. Based on C classes cross similarity $S_{kj} \in \mathbf{R}^{B_u \times C}$, the cross-view prototypical prediction belongs to cth class can be calculated as

$$p_{ko}^c = \frac{e^{s_{kj}^c}}{\sum_c e^{s_{kj}^c}} \in \mathbf{R}^{B_u}. \tag{5}$$

By taking all classes into consideration, we can eventually obtain the cross-view probability prediction $p_{ko} \in \mathbf{R}^{B_u \times C}$ as well. The overall calculation procedure for self-view and cross-view prototypical prediction is shown in the multiview prototypical learning of Fig. 1.

In the context of hyperspectral CD, self-view is considered the internal semantic consistency of a sample pair under varying augmentation states, while cross-view refers to the alignment of feature distributions across distinct perturbation levels. Our prototypical prediction maintains these consistencies through a nonparametric similarity-based response mechanism instead of an auxiliary parametric classifier. Specifically, the class-specific probability is derived by computing the correspondence between patch-level features and batch-wide prototypes. This integrated design ensures that the prototypical representation is self-adaptive and dynamically driven by the feature distribution within each batch, which effectively bridges the gap between weak and strong representations for robust land-cover transformation detection.

B. Multiview Pretraining

To enhance the feature interactions of various perturbed information from different views while maintaining disagreement within a semi-supervised consistency training framework, we propose the MVPL, which synergistically reinforces the weak-to-strong view consistency constraint (WSVCC), where MVPL contains the self-view prototypical consistency constraint (SVPCC) and the cross-view prototypical consistency constraint (CVPCC). The SVPCC benefits from the self-view prototypical prediction $p_{kk} \in \mathbf{R}^{B_u \times C}$, and the CVPCC benefits from the cross-view probability prediction $p_{ko} \in \mathbf{R}^{B_u \times C}$, while the WSVCC is calculated using the model prediction probability $p_k \in \mathbf{R}^{B_u \times C}$.

1) Self-View Prototypical Consistency Constraint: To boost the intraview compactness of the hyperspectral CD prediction probability, we propose the SVPCC method, which leverages $p_{kk} \in \mathbf{R}^{B_u \times C}$ as implicit pseudo label supervision. Therefore, the loss function of SVPCC based on the two-norm is formulated as

$$L_{SV}(k) = \frac{1}{C \times B_u} \|p_{kk} - p_k\|_2 \tag{6}$$

where k is still represented for the sequence number of views.

2) Cross-View Prototypical Consistency Constraint: To obtain dependable knowledge from other training views, we propose the CVPCC, which leverages the cross-view probability prediction $p_{ko} \in \mathbf{R}^{B_u \times C}$ as the implicit pseudo label supervision for $p_k \in \mathbf{R}^{B_u \times C}$. In the calculation of the CVPCC, we introduce the self-view prototypical prediction $p_{kk} \in \mathbf{R}^{B_u \times C}$ as well to adjust the cross-view influence and further enhance the reliability of CVPCC. Specifically, we calculate the maximum value of p_{kk} along class c, termed as the self-view confidence weight $w_k(i)$

$$w_k(i) = \max_c p_{kk}^c(i) \in \mathbf{R}. \tag{7}$$

We can obtain the confidence weights $w_k \in \mathbf{R}^{B_u}$. Then, the optimized loss function of CVPCC can be defined as

$$L_{CV}(k) = \frac{1}{C \times B_u} \|w_k \cdot (p_{ko} - p_k)\|_2 . \tag{8}$$

Therefore, the loss function for the kth view in the MVPL framework can be represented as the sum of the SVPCC loss and the CVPCC loss

$$L_{MVPL}(k) = L_{SV}(k) + L_{CV}(k) . \tag{9}$$

3) Weak-to-Strong View Consistency Constraint: In our unsupervised consistency constraint designation, we also leverage the classic theory of the weak-to-strong consistency principle. To better elucidate the effects of weak-to-strong consistency, we define $p_{uw} = p_1$ and $p_{us} = p_2$, which represent the model's probability predictions corresponding to the two mentioned views. Based on this principle, the probability p_{us} should remain consistent with the probability p_{uw}, which can be regarded as unilateral prediction supervision. Furthermore, it is necessary to introduce a filter τ to exclude p_{uw} with very low probability predictions. Therefore, the loss calculation of WSVCC can be represented as

$$L_{WSV} = \frac{1}{B_u} \sum \mathbb{I}\left(\max\left(\boldsymbol{p}_{uw} > \tau\right)\right) H\left(\boldsymbol{p}_{uw}, \boldsymbol{p}_{us}\right) \tag{10}$$

where the $H(\cdot)$ indicates the entropy between two probabilities. Which can be defined as

$$H\left(\boldsymbol{p}_{us}, \boldsymbol{p}_{uw}\right) = -\sum_{i=1}^{c} \boldsymbol{p}_{us}^{i} \log\left(\boldsymbol{p}_{uw}^{i}\right) . \tag{11}$$

Based on these consistency constraints, we can eventually define the multiview unsupervised consistency constraint loss

$$L_u = \lambda_p \cdot \sum_k L_{MVPL}(k) + \left(1 - \lambda_p\right) \cdot L_{WSV} \tag{12}$$

where λ_p represents the coefficient for prototypical consistent learning and weak-to-strong consistent learning.

4) Loss Function for Labeled Samples: The other training branch for the pretraining step is arranged in a supervised manner. Labeled samples without perturbations are trained in a model for feature extraction directly. Therefore, the supervision loss can be defined as

$$L_s = \frac{1}{B_s} \sum H(p_l, c) \tag{13}$$

where B_s is the batch size for labeled samples, p_l indicates the probability prediction, and c is the corresponding one-hot ground truth.

Based on the mentioned L_u and L_s, the total loss can be formulated along with the unsupervised consistency constraint coefficient λ_u as follows:

$$L = L_s + \lambda_u \cdot L_u. \tag{14}$$

It is worth noting that the perturbed samples are produced via an online process during each training iteration. For each input patch in the selected baseline, the model generates exactly one weakly perturbed view and one strongly perturbed view. Consequently, the total volume of processed samples is dynamically maintained at twice the size of the current pretraining baseline, ensuring that the diversity of the augmented data scales with the sample requirements discussed in our ablation studies.

C. SECA Module

The overall design of the SECA module is illustrated in Fig. 2. Note that SECA is designed as a plug-and-play module and integrated in the backbone from [45]. The first level of semantic explaining information pertains to the logical land cover change types. The core philosophy of it is to perform unsupervised semantic disentanglement by projecting deep features into a structured latent space. Given the prior knowledge that changes in land spatial–spectral information represent a "from-to" problem, we project the deep features onto specific feature clustering channels before performing change classification, assuming there are m different land cover types. Consequently, the feature clustering channels encompass $A_m^2 = m!/((m-2)!)$ change types and one unchanged type. Under this structural constraint, the SECA module effectively functions as a physical-meaningful bottleneck; it forces the high-dimensional spectral–temporal signals to be regularized into specific channels that correspond to distinct land-cover evolution patterns. Thus, the total number of channels allocated to SECA can be designed as $A_m^2 + 1$, and project features can be expressed as $x_{proj} \in \mathbf{R}^{B \times (A_m^2+1) \times H \times W}$. Crucially, the SECA module adaptively populates these semantic channels during training, where the most significant spectral evolution patterns are self-organized into distinct dimensions. This eliminates the need for manual "from-to" labeling while ensuring that the model captures the intrinsic categorical transitions of the hyperspectral data in a label-efficient manner.

The second level of semantic explaining information is concerning to the spatial distance to the central pixel spectrum. Assuming the size of the feature map is $H \times W$, first, we sparsify the matrix indexing coordinates, which means there should be intervals between feature map points, as illustrated in Fig. 3. The new indexing coordinate changes to be size of $(2H-1) \times (2W-1)$. By doing so, the distance measurements can adapt to various shapes of feature maps.

We introduce a conception of logical location that is based on the sparse indexing coordinate. The logical central location can be defined as

$$i_c = \lfloor (2W-1)/2 \rfloor + 1 \tag{15}$$

$$j_c = \lfloor (2H-1)/2 \rfloor + 1. \tag{16}$$

Then, the new logical location for the previous feature point at location i, j is obtained

$$i_l = 2i - 1 \tag{17}$$

$$j_l = 2j - 1. \tag{18}$$

Based on the logical location indexes, we can define the weighting mask

$$w_{i,j} = \begin{cases} 1, & \text{if, } i_l = i_c, j_l = j_c \\ \dfrac{1}{\left(\lfloor \sqrt{(i_l - i_c)^2 + (j_l - j_c)^2} \rfloor\right)^2}, & \text{if, } i_l \neq i_c, j_l \neq j_c. \end{cases} \tag{19}$$

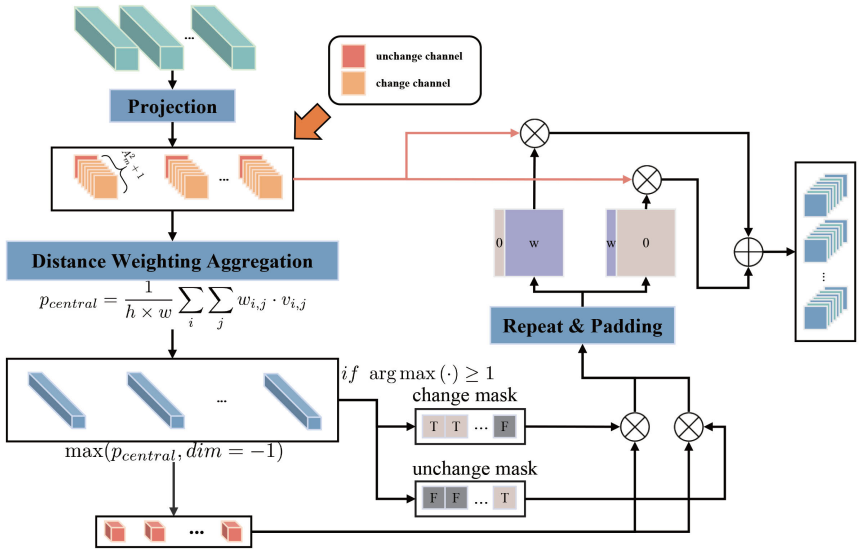


Fig. 2. Overall design of SECA module.

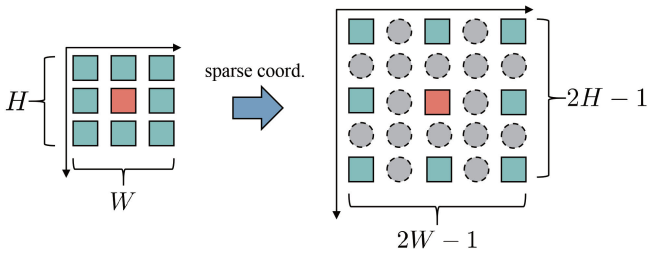


Fig. 3. Construction of sparse coordinate.

For example, if the feature maps input in the SECA module is 3×3, then the weighting mask can be calculated as

$$\begin{Bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 1 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{Bmatrix}. \tag{20}$$

Based on the weighting mask, we can reshape the bth feature map in the spatial dimension into a single aggregation vector

$$p_{central}(b) = F_{SoftMax}\left(\frac{1}{H \times W} \sum_i \sum_j w_{i,j} \cdot value_{i,j}\right) \in \mathbf{R}^{A_m^2+1} \tag{21}$$

where $value_{i,j}$ indicates the feature map value in the location of i, j. Then we find out the corresponding feature channel of the max value based on $p_{central} \in \mathbf{R}^{B \times (A_m^2+1)}$ as the change type, where B indicates the batch size. We define that a location index $\arg\max(p_{central})$ greater than or equal to 1 indicates significant features as "changed features," while an index less than 1 indicates "unchanged features." During the training process, the feature distribution will be automatically constrained to the shape we define. Thus, the boolean mask for "changed features" and "unchanged features" can be represented as follows:

$$M_{ch} = \arg\max(P_{central}) \geq 1 \in \mathbf{R}^B \tag{22}$$

$$M_{un} = \neg M_{ch} \in \mathbf{R}^B. \tag{23}$$

And the corresponding maximum value for all feature samples in the batch is defined as

$$V_{max} = \max(p_{central}) \in \mathbf{R}^B. \tag{24}$$

To achieve the distance and logical land cover change weighting map, the padding function $F_{padding}(\cdot)$ and the repeating function $F_{repeat}(\cdot)$ are leveraged to expand V_{max} that is filtered by both M_{ch} and M_{un}

$$W_{ch} = F_{padding}\left(F_{repeat}(M_{ch} \odot V_{max})\right) \in \mathbf{R}^{B \times (A_m^2+1)} \tag{25}$$

$$W_{un} = F_{padding}\left(F_{repeat}(M_{un} \odot V_{max})\right) \in \mathbf{R}^{B \times (A_m^2+1)} \tag{26}$$

where $\odot$ indicates the point-wise production. As shown in Fig. 2, take $W_{ch} \in \mathbf{R}^{B \times (A_m^2+1)}$ as an example, the first channel is filled with zeros to prevent influence from the corresponding unchanged channel in V_{max}. As for the remaining A_m^2 channels, the maximum values, which we assume fall within the range of the changed feature channels, are repeated across all changed channels. The calculation process of W_{un} is similar.

Eventually, we can simply use both W_{ch} and W_{un} as the weighting maps for the projected deep features $x_{proj} \in \mathbf{R}^{B \times (A_m^2+1) \times H \times W}$ and fuse their feature outputs

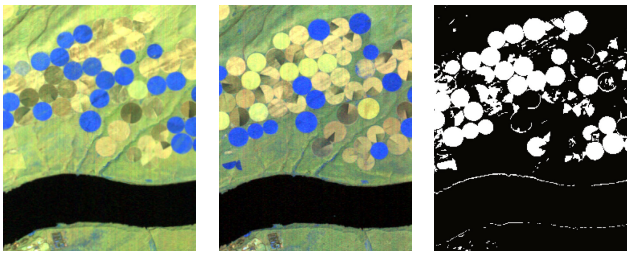
$$x_{out} = W_{ch} \odot x_{proj} \oplus W_{un} \odot x_{proj} \in x_{proj} \in \mathbf{R}^{B \times (A_m^2+1) \times H \times W} \tag{27}$$

where $\oplus$ implies point-wise addition in the channel dimension.

The SECA method introduces a channel-level feature constraint mechanism that automatically clusters different types of changes into distinct semantic channels. Although the core task remains binary CD, the model achieves a more structured representation of change features, enhancing the semantic differentiation of the outputs. This allows the model to distinguish

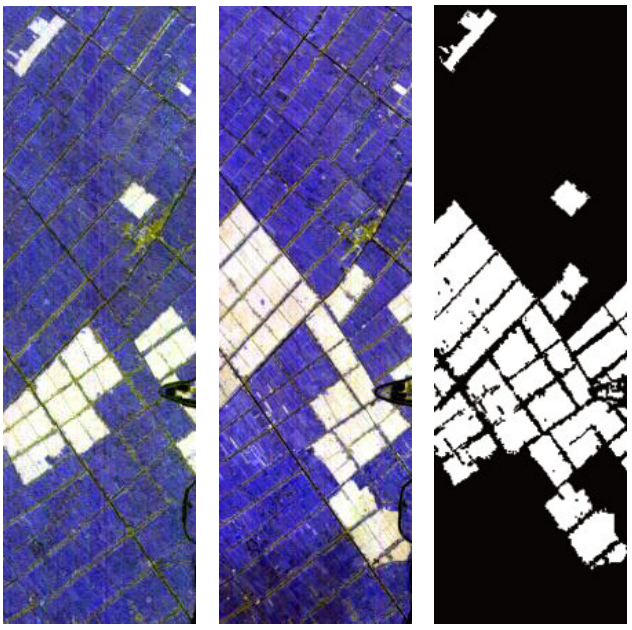
TABLE I
DETAIL OF THREE PUBLIC DATASETS

Dataset	Spatial Size	Band	Date1	Date2	Training Rate	Training Samples
Irrigated	307×241	156	May 1st, 2004	May 8th, 2007	9.7%	7250
Wetland	450×140	156	May 3rd, 2006	Apr. 23rd, 2007	9.7%	6173
River	463×241	198	May 3rd, 2013	Dec. 31st, 2013	4.03%	4500



(a) (b) (c)

Fig. 4. Irrigated agricultural dataset with false-color map (band 134, 90, and 75 as RGB). (a) USA farmland image captured on May 1, 2004. (b) USA farmland image captured on May 8, 2007. (c) Binary ground truth for the irrigated agricultural dataset.



(a) (b) (c)

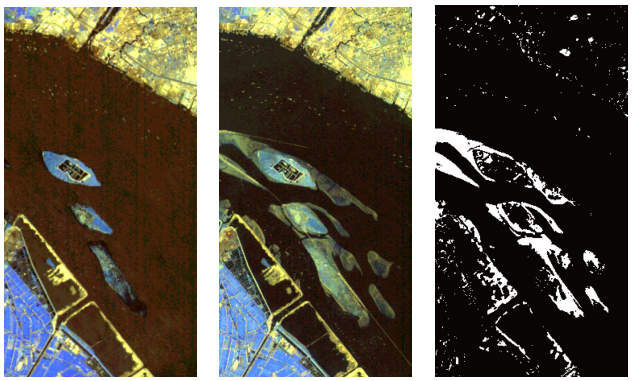
Fig. 5. Wetland agricultural dataset with false-color map (band 134, 90, and 75 as RGB). (a) China's farmland captured on May 3, 2006. (b) China's farmland captured on April 23, 2007. (c) Binary ground truth for the wetland agricultural dataset.

between various types of changes even without explicit land cover labels, providing more interpretable information for downstream geospatial analysis.

IV. EXPERIMENTS

A. Description of Datasets

This article evaluates the performance of the MVPL framework integrated with SETSST on three publicly available multitemporal hyperspectral CD datasets from the EO-1 satellite. The Hyperion sensor provides spectral data from 0.4 to 2.5 μm with 242 bands and a spatial resolution of 30 m. Spectral bands with low signal-to-noise ratios were excluded, and binary ground truth maps were generated via visual analysis and on-site inspections. The irrigated agricultural dataset covers Hermiston city, Oregon, with images from May 2004 and May 2007, including 307×241 pixels and 156 bands [46]. The wetland agricultural dataset[1] represents farmland in Zhejiang Province, China, with images from May 2006 and April 2007, also with 156 bands [46]. The river dataset,[2] from Jiangsu Province, China, includes images from May and December 2013 with 198 bands [19]. Stratified random sampling and multiple experiments (5 times) were applied for comparisons. Following their original work and maintaining sample consistency, the benchmark training rate is 9.7% for the Irrigated and Wetland datasets, and the unchanged-to-changed sample ratio for the river dataset is kept at 2:1 to address class imbalance. Ground truth and false-color maps are shown in Figs. 4–6, with dataset details in Table I.



(a) (b) (c)

Fig. 6. River dataset with false-color map (band 134, 90, and 75 as RGB). (a) China River captured on May 3, 2013. (b) China River captured on December 31, 2013. (c) Binary ground truth for the river dataset.

B. Comparison Methods and Experimental Settings

To validate our method, we compare its performance with traditional hyperspectral CD techniques and advanced deep learning approaches. Conventional methods include CVA (2007) [10], which uses difference maps and unsupervised thresholding, and SVM (2006) [11], commonly applied for land cover CD. Among the deep learning approaches, we evaluate our model against GETNET (2019), which is built on the LSConvolution framework [19]. Since our approach incorporates learning in both spectral and spatial domains, we

[1]Irrigated and wetland datasets can be downloaded from http://rslab.ut.ac.ir
[2]River dataset can be downloaded from http://crabwq.github.io

TABLE II
MODEL COMPARISON RESULTS ON IRRIGATED AGRICULTURAL DATASET (REPEAT 5 TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9286	0.7704	0.8127	0.8127
SVM	0.9462±0.0094	0.8233±0.0305	0.8558±0.0233	0.8911±0.0170
GETNET	0.9496±0.0031	0.8510±0.0133	0.8610±0.0079	0.8939±0.0058
WCRN	0.9420±0.0031	0.8308±0.0119	0.8412±0.0076	0.8790±0.0055
ChangeFormer	0.8736±0.0170	0.7874±0.0756	0.6080±0.0863	0.6831±0.0854
TFR	0.9709±0.0013	0.9217±0.0035	0.9181±0.0005	0.9368±0.0026
SSTFormer	0.9601±0.0054	0.8850±0.0325	0.8892±0.0134	0.9151±0.0098
GLAFormer	0.9457±0.0040	0.8335±0.0195	0.8518±0.0090	0.8874±0.0062
DA-Former	0.7718±0.1154	0.5252±0.1299	0.5092±0.1639	0.6528±0.1000
No MVPL	0.9717±0.0011	0.9157±0.0059	0.9206±0.0030	0.9389±0.0023
Proposed	**0.9768±0.0018**	**0.9421±0.0109**	**0.9341±0.0049**	**0.9490±0.0038**

also include WCRN (2022) as a comparative method for two-domain learning, implemented using a CNN architecture [21]. For Transformer-based models, we consider ChangeFormer (2023) [47] and SSTFormer (2023) [24]. In addition, we include our previous work, TFR (2023), which integrates an attention mechanism-based spatial–spectral module [41]. We also make a comparison with recent methods, which are GLAFormer and DA-Former [48], [49]. The default parameter settings from the respective studies are used for all the deep learning-based methods.

The algorithm implementations are based on the PyTorch platform. Due to the model's flexible architecture, it supports inputs of varying patch sizes without requiring a fixed spatial resolution. The Adagrad optimizer is used throughout the process, with a learning rate of 1×10^{-4}. The cross-entropy loss function is selected to compute the fine-tuning loss.

C. Evaluation Metrics

To ensure a fair and comprehensive performance comparison across the three hyperspectral CD datasets, we utilize several evaluation metrics: overall accuracy (OA), changed accuracy (CA), Kappa coefficient (Kappa), and $F1$-score. Higher values in these metrics reflect superior model performance.

Overall Accuracy: For pixel-level classification tasks, accuracy serves as a straightforward and effective metric to evaluate model performance. The OA is determined using the following formula:

$$OA = \frac{TP + TN}{TP + TN + FP + FN}. \tag{28}$$

Changed Accuracy: CA, also referred to as precision, is used to evaluate the model's ability to detect changed areas. It represents the ratio of true positives to the total number of true positives and false positives. The formula is as follows:

$$CA = \frac{TP}{TP + FP}. \tag{29}$$

Kappa Coefficient (Kappa): The Kappa coefficient is a widely used metric for patch-level classification that accounts for class imbalances, offering a more balanced evaluation of model performance across different datasets. The formula for calculating Kappa is as follows:

$$Kappa = \frac{OA - PE}{1 - PE} \tag{30}$$

$$PE = \frac{(TP + FP)(TP + FN) + (TN + FP)(TN + FN)}{(TP + TN + FP + FN)^2}. \tag{31}$$

F1-Score: The $F1$-score is a combined metric used in binary classification tasks, balancing both precision and recall. It evaluates the model's effectiveness by accounting for true positives, false positives, and false negatives, providing a comprehensive measure of performance. The $F1$-score ranges from 0 to 1, with higher values indicating better performance. This metric is especially useful for imbalanced datasets

$$F1 - Score = \frac{2 \times CA \times RE}{CA + RE} \tag{32}$$

$$RE = \frac{TP}{TP + FN}. \tag{33}$$

D. Comparison Experiments

1) Experiments on the Irrigated Agricultural Dataset: The comparison results for the irrigated agricultural dataset are shown in Table II. Our proposed SETSST with MVPL pretraining achieves the highest performance across all metrics, including OA, CA, Kappa, and $F1$-score. TFR is the closest competitor, with an OA of 0.9709. Traditional methods like CVA and SVM achieve OAs of 0.9286 and 0.9462, respectively, highlighting their limitations in capturing temporal–spatial–spectral information. Convolutional models, GETNET and WCRN, achieve OAs of 0.9496 and 0.9420. ChangeFormer, designed for multispectral data, delivers the lowest OA of 0.8736, struggling with spectral features. SST-Former, tailored for HSIs, ranks second-best with an OA of 0.9601. DA-Former, as a recent supervised method, cannot detect change regions effectively, which achieves an OA of 0.7718. Even without MVPL pretraining, our model achieves the best performance, demonstrating its robustness in fully supervised scenarios.

The CD result comparison maps for the irrigated agricultural dataset are shown in Fig. 7. The MVPL framework, combined with SETSST, produces results most similar to the ground truth, effectively capturing subtle change details, such as agricultural land and river boundaries. Key differences are marked with red boxes. CVA misses farmland areas and omits fringe details, while SVM mistakenly detects background information. GETNET and WCRN fail to capture river boundary details. ChangeFormer performs the worst, merging all agricultural land areas and losing subtle details. TFR and SST-Former are the best competitors, but their main differences lie in circular farmland boundaries. Without MVPL-based pretraining, the results show omissions near the river boundary.

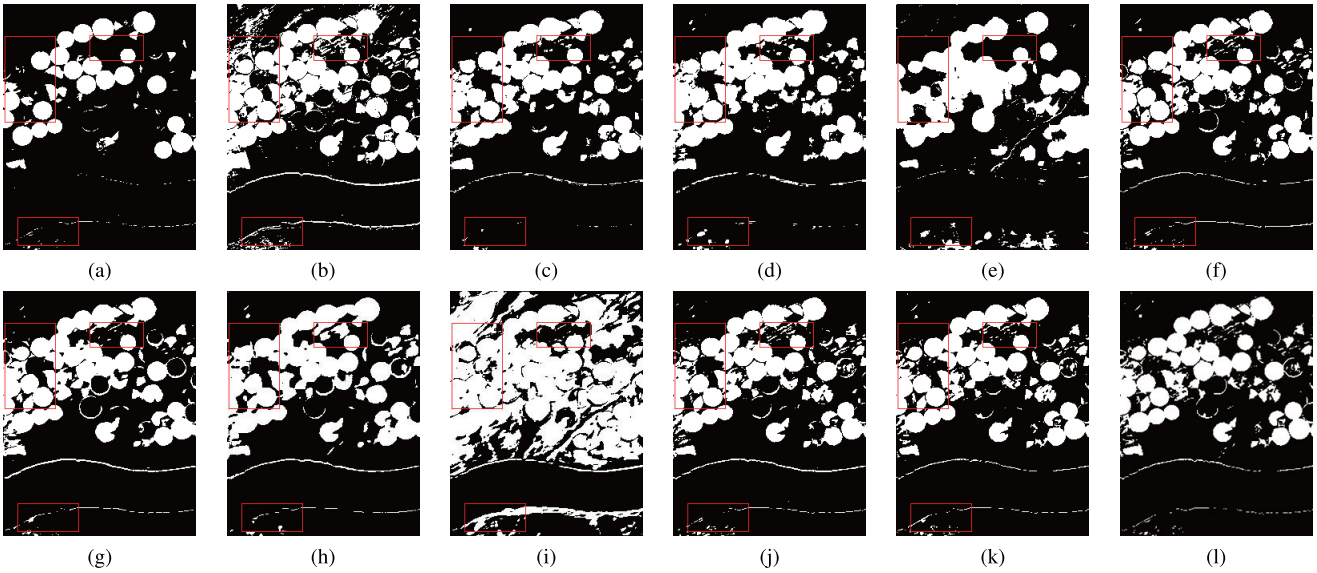


Fig. 7. Change map results of different methods on the irrigated agricultural dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) GLAFormer. (i) DA-Former. (j) Without MVPL. (k) Proposed. (l) Ground truth.

TABLE III

MODEL COMPARISON RESULTS ON WETLAND AGRICULTURAL DATASET (REPEAT 5 TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9525	0.9032	0.8859	0.9196
SVM	0.9497±0.0337	0.9088±0.0130	0.8748±0.0917	0.9092±0.0706
GETNET	0.9611±0.0037	0.8954±0.0133	0.9082±0.0082	0.9360±0.0054
WCRN	0.9602±0.0008	0.9045±0.0062	0.9054±0.0017	0.9337±0.0011
ChangeFormer	0.8636±0.0443	0.7646±0.0621	0.6651±0.1157	0.7597±0.0864
TFR	0.9750±0.0101	0.9443±0.0367	0.9402±0.0229	0.9580±0.0155
SSTFormer	0.9743±0.0033	0.9360±0.0051	0.9384±0.0098	0.9566±0.0067
GLAFormer	0.9652±0.0082	0.9131±0.0339	0.9174±0.0184	0.9422±0.0122
DA-Former	0.8186±0.0826	0.6320±0.1132	0.6334±0.1398	0.7660±0.0800
No MVPL	0.981±0.0004	0.9564±0.0011	0.9542±0.0010	0.9676±0.0007
Proposed	**0.9831±0.0005**	**0.9694±0.0042**	**0.9592±0.0014**	**0.9710±0.0010**

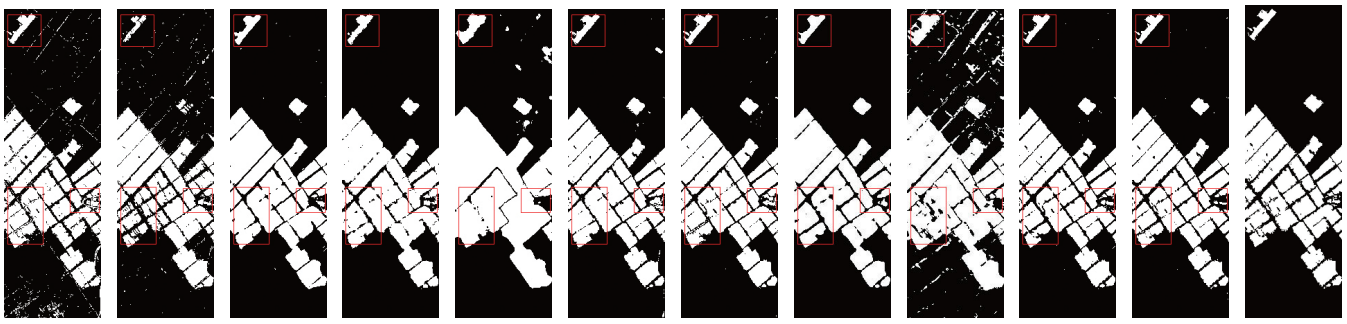


Fig. 8. Change map results of different methods on the wetland agricultural dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) GLAFormer. (i) DA-Former. (j) Without MVPL. (k) Proposed. (l) Ground truth.

2) Experiments on the Wetland Agricultural Dataset: The model comparison results for the wetland agricultural dataset are presented in Table III. The proposed SETSST with MVPL pretraining outperforms all comparison methods across metrics, including OA, CA, Kappa, and $F1$-score. Even without MVPL, SETSST achieves strong performance. Traditional methods, CVA and SVM, achieve OA scores of 0.9525 and 0.9497 but fall short of capturing farmland changes effectively. TFR is the closest competitor, with an OA slightly lower than SETSST, while SST-Former performs similarly, achieving an OA of 0.9743. GETNET delivers acceptable results with an OA of 0.9611 but has poor CA performance. WCRN shows lower OA (0.9602) and Kappa (0.9054). ChangeFormer experiences the steepest drop, with an OA of 0.8636 and high variability, suggesting it struggles to handle the transition from multispectral to hyperspectral data.

Fig. 8 shows the change maps generated by various methods for the wetland agricultural dataset. The proposed STSST, combined with the MVPL pretraining framework, produces change maps that closely resemble the ground truth. The

TABLE IV
MODEL COMPARISON RESULTS ON RIVER DATASET (REPEAT 5 TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9280	0.5492	0.6617	0.6992
SVM	0.9147±0.0393	0.5431±0.1098	0.5838±0.0998	0.6275±0.0844
GETNET	0.9352±0.0099	0.5829±0.0424	0.6821±0.0339	0.7163±0.0292
WCRN	0.9310±0.0061	0.5648±0.0249	0.6637±0.0181	0.7000±0.0154
ChangeFormer	0.9153±0.0189	0.5287±0.0908	0.5519±0.0523	0.5971±0.0430
TFR	0.9488±0.0084	0.6450±0.0477	0.7340±0.0336	0.7615±0.0293
SSTFormer	0.9061±0.0297	0.4955±0.0769	0.5929±0.0731	0.6396±0.0613
GLAFormer	0.9443±0.0142	0.6288±0.0739	0.7134±0.0518	0.7432±0.0444
DA-Former	0.9044±0.0284	0.5958±0.2744	0.2540±0.2214	0.2833±0.2529
No MVPL	0.9562±0.0037	0.6790±0.0228	0.7661±0.0160	0.7898±0.0141
Proposed	**0.9653±0.0013**	**0.7558±0.0162**	**0.7980±0.0057**	**0.8170±0.0051**

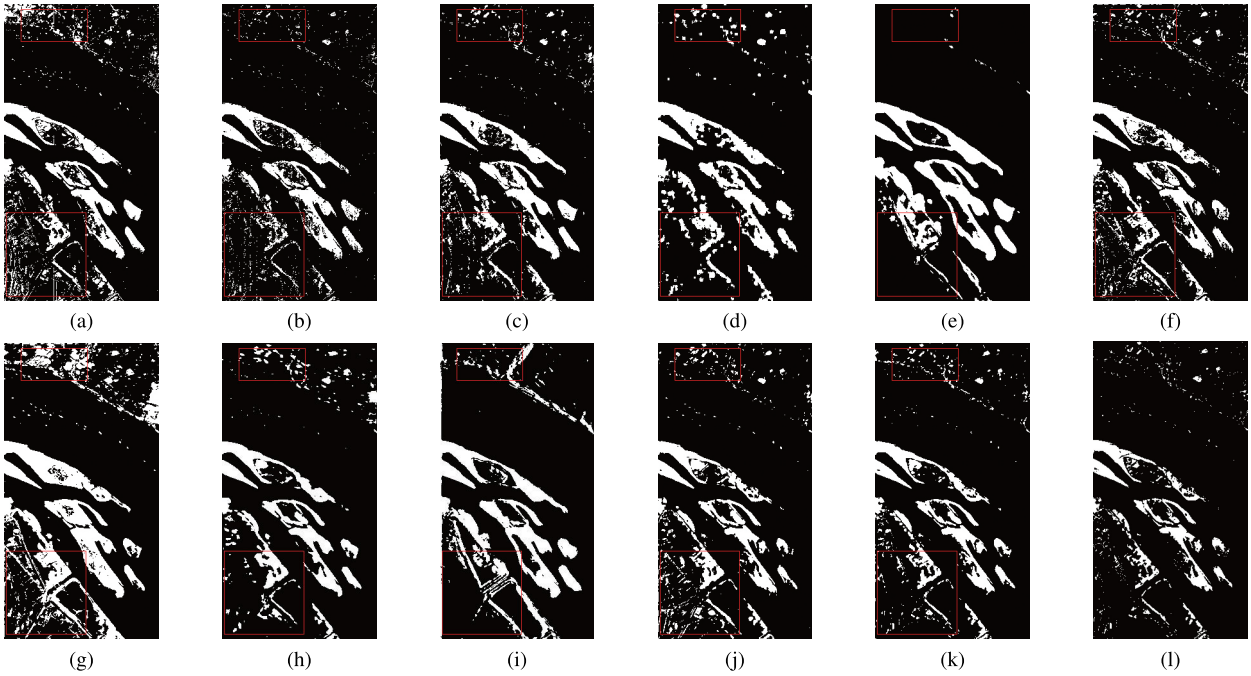


Fig. 9. Change map results of different methods on the river dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) GLAFormer. (i) DA-Former. (j) Without MVPL. (k) Proposed. (l) Ground truth.

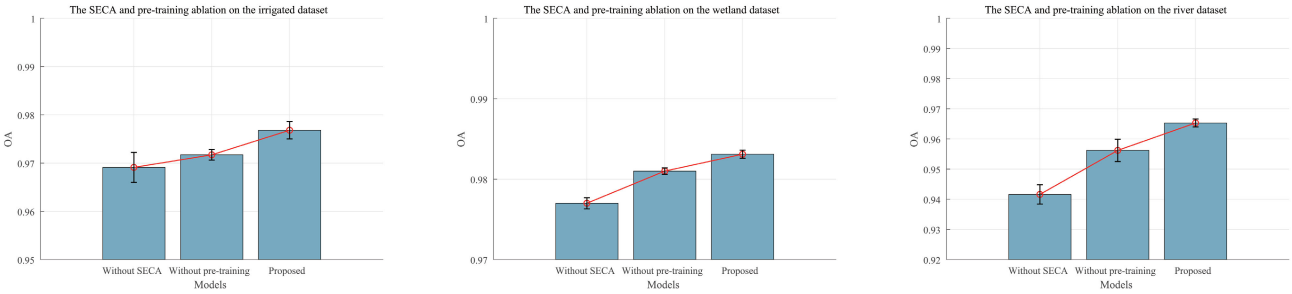
predicted map effectively identifies subtle changes in farmland blocks, while also clearly distinguishing unchanged areas. Key differences in each change map are highlighted with red boxes. The conventional methods, CVA and SVM, incorrectly detect multiple boundary lines between farmland areas, resulting in numerous noise-like artifacts on the upper side of the CD maps. The visual results produced by ChangeFormer and two-domain WCRN are quite similar, with both models being unable to capture any boundary change information within the farmland areas. TFR, SST-Former, and our proposed model without MVPL pretraining show the closest visual similarity to the ground truth map among the comparison methods, successfully preserving subtle change details. However, some background information is mistakenly detected as change areas. The semi-supervised DA-Former exhibits a significant amount of background noise, which disturbs the visual effects.

3) Experiments on the River Dataset: Table IV presents the comparison results of different methods on the river dataset. The proposed SETSST with the MVPL-based pretraining framework outperforms all others in OA, CA, Kappa, and $F1$-score. Even without the MVPL framework, the SETSST still achieves the best OA performance under standard training conditions. Among the comparison models, TFR is the strongest competitor to our approach, with an OA of 0.9488, a CA of 0.6459, a Kappa of 0.7340, and an $F1$-score of 0.7615. GETNET is the second-best performer, achieving an OA of 0.9352, demonstrating its effectiveness on datasets with linear spectral relationships and local spatial texture information. As Transformer-based methods, SST-Former and ChangeFormer deliver less than ideal performances, with OAs of 0.9061 and 0.9153, respectively. These suboptimal results indicate that both models have difficulty handling long-range similarities in land cover attributes within the river area. The following model, WCRN, achieves an OA of 0.9310, a CA of 0.5648, a Kappa of 0.6637, and an $F1$-score of 0.7. Regarding traditional methods, CVA and SVM show comparable OA performances, with scores of 0.928 and 0.9147, respectively.

Fig. 9 presents the predicted CD maps for the river dataset from various methods. The SETSST, combined with the MVPL-based pretraining framework, closely aligns with the ground truth, showcasing superior performance. Key

TABLE V
SECA AND MULTIVIEW PRETRAINING STEP ANALYSIS OF THE MODEL

Dataset	SECA	Pre-training	OA	CA	Kappa	F1-Score
I.A. Dataset	✗	✗	0.9691	0.9088	0.9134	0.9335
	✓	✗	0.9717	0.9157	0.9206	0.9389
	✓	✓	**0.9768**	**0.9421**	**0.9341**	**0.9490**
W.A. Dataset	✗	✗	0.9770	0.9459	0.9449	0.9611
	✓	✗	0.9810	0.9564	0.9542	0.9676
	✓	✓	**0.9831**	**0.9694**	**0.9592**	**0.9710**
River Dataset	✗	✗	0.9416	0.6065	0.7052	0.7363
	✓	✗	0.9562	0.6790	0.7661	0.7898
	✓	✓	**0.9653**	**0.7558**	**0.7980**	**0.8170**



(a) (b) (c)

Fig. 10. SECA and multiview pretraining step ablation model performance on three datasets. (a) Irrigated agricultural dataset. (b) Wetland agricultural dataset. (c) River dataset.

differences are marked with red boxes. CVA generates a map with errors in the bottom red box, while SVM introduces noise-like errors. Among deep learning methods, GETNET produces similar visuals but misidentifies edges in the bottom-left red box. WCRN detects plaque-shaped changes, losing significant details. Transformer-based methods perform variably: ChangeFormer fails to capture intricate details, and SST-Former detects excessive incorrect changes. TFR offers results closest to the ground truth, with only minor omissions compared to SETSST. DA-Former provides only a coarse visual delineation of CD, with significant loss of fine-scale variation details and misdetected changes in the red-boxed regions.

TABLE VI
COMPREHENSIVE RESULTS OF MULTIVIEW UNSUPERVISED CONSISTENCY CONSTRAINTS ABLATION EXPERIMENTS

Dataset	MVPL	WSVCC	OA
I.A. Dataset	✗	✗	0.9717
	✓	✗	0.9730
	✗	✓	0.9742
	✓	✓	**0.9768**
W.A. Dataset	✗	✗	0.9810
	✓	✗	0.9829
	✗	✓	0.9826
	✓	✓	**0.9831**
River Dataset	✗	✗	0.9562
	✓	✗	0.9629
	✗	✓	0.9632
	✓	✓	**0.9653**

E. Ablation Study

1) SECA Module Ablation: To validate the SECA module's effectiveness, we conducted ablation experiments, removing it and the pretraining step, and evaluated performance on three public datasets (Table V). After removing the SECA module and pretraining step, the proposed deep model is trained in a fully supervised manner, achieving an OA of 0.9691 on the irrigated agricultural dataset. However, its performance is lower compared to the deep model enhanced with the SECA module. When the ablation experiment is carried out on the wetland agricultural dataset, the model without the SECA module only achieves OA of 0.9770, which is inferior to that with the SECA module. A similar trend is observed on the river dataset, where the model with the ablation module achieves an OA of 0.9416. To visually illustrate the performance improvement among the ablation models, a statistical chart is presented in Fig. 10. The CD performance improves as both the SECA and the multiview pretraining step are integrated across all three public datasets.

The visual detection results of the SECA module ablation are shown in Fig. 10(a)–(c), with differences marked in red boxes. In Fig. 10(a), the main differences occur near the circular farmland, where the model without SECA and multiview pretraining shows the least accurate detections. On the wetland dataset [Fig. 10(b)], the differences appear at farmland boundaries, with the SECA ablation model showing the most incorrect results. Fig. 10(c) for the river dataset shows that the SECA ablation model fails to identify change areas, while the multiview pretraining step ablation model mistakenly detects more change areas. The above ablation experimental results demonstrate the effectiveness of the proposed SECA module.

2) Multiview Prototype Learning Ablation: The MVPL framework aims to effectively leverage aggregation features

TABLE VII
COMPARISON OF SUPERVISED METHODS BEFORE AND AFTER THE MULTIVIEW PRETRAINING

Dataset	Methods	Before	After
I.A. Dataset	GETNET	0.9496	0.8743
	WCRN	0.9420	0.8399
	ChangeFormer	0.8736	**0.9475**
	TFR	0.9709	**0.9716**
	SST-Former	0.9601	0.9535
	GLAFormer	0.9457	0.9434
W.A. Dataset	GETNET	0.9611	0.9221
	WCRN	0.9602	0.9356
	ChangeFormer	0.8636	**0.9698**
	TFR	0.9750	**0.9776**
	SST-Former	0.9743	**0.9745**
	GLAFormer	0.9652	**0.9673**
River Dataset	GETNET	0.9352	0.9042
	WCRN	0.9310	0.8808
	ChangeFormer	0.9153	**0.9344**
	TFR	0.9488	**0.9507**
	SST-Former	0.9061	**0.9250**
	GLAFormer	0.9443	**0.9457**

and consistency features from both labeled and unlabeled samples. Therefore, based on the loss function construction, we design MVPL ablation experiments that contain the MVPL loss function ablation and the WSVCC loss function ablation on three public datasets. The comprehensive results are shown in Table VI.

As shown in Table VI, the model incorporating both the MVPL and WSVCC loss functions delivers the best performance across all three datasets. Conversely, removing both loss functions results in the poorest detection outcomes across all datasets. We also perform ablation experiments by removing each component individually, and the results demonstrate that both consistency constraints for prototypes and original features are equally essential.

The ablation results indicate that the proposed multiview pretraining is sensitive to the underlying architecture, Table VII. While it achieves significant gains on Transformer-based models like ChangeFormer, performance on CNN-based methods (e.g., GETNET and WCRN) is less optimal. This is primarily because the mandatory structural modifications required to extract latent features from these supervised models, combined with the use of fixed hyperparameters optimized for our own framework, may not align with the localized inductive bias of convolutional layers. Nonetheless, the consistent performance across the Transformer series validates the fundamental efficacy of the prototype learning mechanism.

3) Hyperparameter Ablation: To validate the ability of the pretraining step with the MVPL framework to effectively leverage features from both labeled and unlabeled samples, we set several parameters according to their practical mathematical significance. In the hyperparameter ablation experiments, we consider training sample rate, confidence hyperparameter τ, prototype balance coefficient λ_p, and unsupervised loss coefficient λ_u to analyze the influence on the proposed method's performance.

The training rate is used to determine the rate of labeled samples included in step 1 of the pretraining process. Fig. 11 presents the training rate analysis, which evaluates the model's sensitivity to varying amounts of supervision. The asterisk-marked 100% sample rate indicates the use of the initial supervised benchmark training samples, which constitute 9.7%, 9.7%, and 4.03% of the total samples. It is important to note that the standard benchmark training sample rate is detailed in the dataset section, aligning with their original supervised works. Consequently, the training rate analysis is based on the product of the ratio and the initial training samples, for instance, $40\% \times 9.7\%$. This progressive reduction analysis demonstrates the superior sample efficiency of our semi-supervised framework under extreme label-limited conditions.

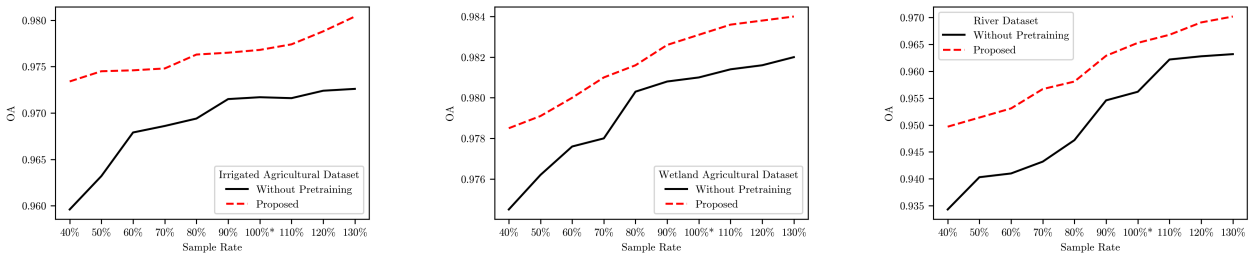
Fig. 11(a) shows the OA performance trend of the SETSST model on the irrigated agricultural dataset, improving as training samples increase. With MVPL-based pretraining, the model significantly outperforms the version without it. On the wetland agricultural dataset, the method steadily improves in OA, with clear differences between the full and ablation models across all sample rates. For the river dataset, the model without pretraining reaches a performance bottleneck at a 110% training rate, maintaining noticeable differences compared to the proposed method at all rates.

The confidence hyperparameter τ is used to filter high-confidence predictions generated under weak perturbation, which plays a crucial role in determining the effectiveness of consistency guidance. The influence by τ on three datasets is shown in Table VIII. When $\tau = 0.6$, the proposed method can achieve the best performance. Therefore, we set the corresponding τ value as the default hyperparameter value.

Due to the loss function in the pretraining step being a nested combination of MVPL, WSVCC, and unsupervised item according to formula from (9) to (14), we conduct hyperparameter ablation of λ_p and λ_u to analyze the influence on the CD performance. The comprehensive results are shown in Tables IX and X. When $\lambda_p = 0.8$ and $\lambda_u = 2$, the proposed method achieves its best CD performance across three public datasets. As a result, we use these optimal hyperparameter values for the loss function during the pretraining step as the default settings.

V. DISCUSSION

The extensive comparison and ablation experiments illustrate the superiority of our proposed MVPL framework, incorporating a general SETSST crossing metrics of OA, CA, Kappa, and $F1$-score. In comparison with conventional methods, both the unsupervised CVA and supervised SVM fail to effectively leverage the rich temporal, spatial, and spectral information, leading to suboptimal detection results. As for deep learning methods, convolution-based models like GETNET and WCRN are designed to extract local spatial–spectral features, but they are unable to fully capture temporal similarities. Transformer-based models like ChangeFormer, TFR, and SST-Former can effectively utilize local temporal–spatial–spectral dependencies. However, they have limitations in capturing semantic land cover attribute information comprehensively. It is noteworthy that models like ChangeFormer, originally designed for image-level remote



(a) (b) (c)

Fig. 11. Sample rate comparison on (a) irrigated agricultural dataset, (b) wetland agricultural dataset, and (c) river dataset.

TABLE VIII

τ INFLUENCE ON THE OA PERFORMANCE

τ	0.5	**0.6**	0.7	0.8	0.9
Irrigated Agriculture	0.9739	**0.9784**	0.9756	0.9740	0.9743
Wetland Agriculture	0.9810	**0.9841**	0.9825	0.9824	0.9820
River Area	0.9586	**0.9653**	0.9597	0.9552	0.9518

TABLE IX

CD PERFORMANCE INFLUENCED BY HYPERPARAMETER λ_p

λ_p	0.2	0.4	0.6	0.8
Irrigated Agriculture	0.9716	0.9736	0.9758	0.9784
Wetland Agriculture	0.9812	0.9818	0.9821	0.9841
River Area	0.9529	0.9543	0.9627	0.9653

TABLE X

CD PERFORMANCE INFLUENCED BY HYPERPARAMETER λ_u

λ_u	0.25	0.5	1	2	4
Irrigated Agriculture	0.9684	0.9737	0.9778	**0.9784**	0.9748
Wetland Agriculture	0.9806	0.9814	0.9825	**0.9841**	0.9824
River Area	0.9572	0.9587	0.9597	**0.9653**	0.9605

sensing tasks, exhibit a paradigm mismatch when applied to the pixel-level or patch-level requirements of HSCD. They often fail to effectively suppress high-dimensional spectral noise or aggregate global semantic cues across a batch.

The proposed MVPL can leverage low-density aggregation representations from both labeled and unlabeled samples. For unlabeled samples, the framework promotes shared feature characteristics and ensures proper feature interaction across perturbations, maintaining disagreement per the weak-to-strong consistency principle. In real-world applications, labeling can be time-consuming, especially in hyperspectral CD. The proposed method achieves high CD accuracy with fewer labeled samples, demonstrating that the MVPL-based framework can reduce the dependency on labeled data.

Despite its effectiveness, the proposed method has certain limitations. Specifically, the two-phase training schedule introduces additional computational expenditure. While the MVPL-based consistency task effectively leverages both labeled and unlabeled data, it inevitably extends the overall training duration. Furthermore, the complex consistency constraints and weak-to-strong prediction alignments contribute to increased computational overhead. Given that prototypes are inherently efficient, future work could further extend the use of aggregated representations throughout the pretext task to better optimize resource utilization.

VI. CONCLUSION

This article introduces a semi-supervised training strategy with MVPL and SETSST models. MVPL employs two feature aggregation prototypes for different perturbations, generating self-view and cross-view prototypical similarity maps. These maps enable the computation of MVPL loss, facilitating unsupervised feature extraction through weak-to-strong prediction alignment. The SETSST model integrates the SECA module to capture semantic information from land cover attributes, guiding deep features into specific clustering channels based on prior knowledge, thereby enhancing the model's explainability.

The proposed method is executed on three publicly available hyperspectral CD datasets. The results show that our proposed method can achieve both visual and quantitative effects, which demonstrate the superior performance when compared to other SOTA methods.

For the value in diverse CD tasks, the MVPL-based STSST can be regarded as a benchmark model with an attention mechanism and can serve as a reference for other semi-supervised strategies, where the pretext task is without manual intervention. In future research, we will concentrate on developing more lightweight pretext tasks to mitigate heavy time and space costs. In addition, in response to the growing demand for big data in hyperspectral remote sensing, further exploration of large hyperspectral models will be pursued.

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